

Predictors of Earnings Risk with Machine Learning

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August 1, 2026

Abstract

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Keywords: machine learning, restricted income profile, earnings instability, risk

JEL Codes: D8, J0, D3

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1 Introduction

2 Theoretical Framework

As mentioned before, the model used is the same model from Drewianka and Oberg (2025) [5]. The model is a standard RIP model where people's income are modeled as a function of characteristics, but where there are no individual level effects. Then each individual has a deviation from the expected income process which is modeled as a shock. For example all 22 year old men from the US will have the same expected income process based off of those characteristics. However their actual income will deviate from this expected income process due to shocks.

$$u_{it} = \pi_{it} + \nu_{it} \quad (1)$$

$$\pi_{it} = \rho \pi_{i(t-1)} + \eta_{it}, \quad (2)$$

u_{it} is the shock or deviation from the expected income process for a given person i in time t . This process of shocks can be decomposed into two types of shocks: persistent and temporary shocks as outlined in equation 1. π_{it} is the cumulative effect from the persistent shocks which are further modeled in equation 2 while ν_{it} are the temporary shocks. The temporary shocks along with η_{it} are assumed to be mean zero and independent across individuals and time. The persistent shocks are further modeled as an autoregressive process with ρ being the persistence of the shocks and η_{it} being the actual shock.

To get a measure of earnings risk the variance of η_{it} across time is calculated. This is capturing the correlation of permanent income shocks across time for each individual. To calculate this the following equation is used:

$$\Omega_i(t, t+k) \equiv u_{i(t+k)} - u_{it} \quad (3)$$

$$\Omega_i(t, t+k) = \nu_{i(t+k)} - \nu_{it} + \sum_{j=1}^k \eta_{i(t+j)}. \quad (4)$$

This omega equation differences u_{it} across a window t to $t+k$. This is then multiplied by another omega function across the window $t-j$ to $t+k+q$. This will result in the persistent shocks in the interval $(t, t+k)$ showing up twice while the other persistent shocks don't necessarily show up twice. This results in the variance of the shocks across time k as follows:

$$E\left[\frac{1}{k} \Omega_i(t, t+k) \Omega_i(t-j, t+k+q)\right] = \sigma_{\eta_i}^2, \quad (5)$$

$$\gamma_{itjq} \equiv \frac{1}{2} \Omega_i(t, t+2) \Omega_i(t-j, t+2+q). \quad (6)$$

This is then further used to get the following moment condition for the variance of η_{it} that is a function of $(j+k+q)$ and j and can be estimated in a regression framework to get a measure of lifetime earnings risk γ_{itjq} . For more on the derivation see Drewianka and Oberg (2025) [5]:

$$E[\gamma_{itjq}] \approx [\delta^2 E\pi_{i(t-j)}^2 (j+k+q)] - [\delta s_i(t-j+1, t) j] + [\sigma_{\eta_i(t+1)}^2 - \delta \sigma_{\eta_i(t+1)}^2 (k+q)] \quad (7)$$

$$E[\gamma_{itjq}] = \sigma_{\eta_i(t+1)}^2 + [\delta^2 E\pi_{i(t-j)}^2 - \delta \sigma_{\eta_i(t+1)}^2] (j+k+q) + \delta [\sigma_{\eta_i(t+1)}^2 - s_i(t-j+1, t)] j \quad (8)$$

3 Data

4 Empirical Strategy

5 Results

6 Conclusion

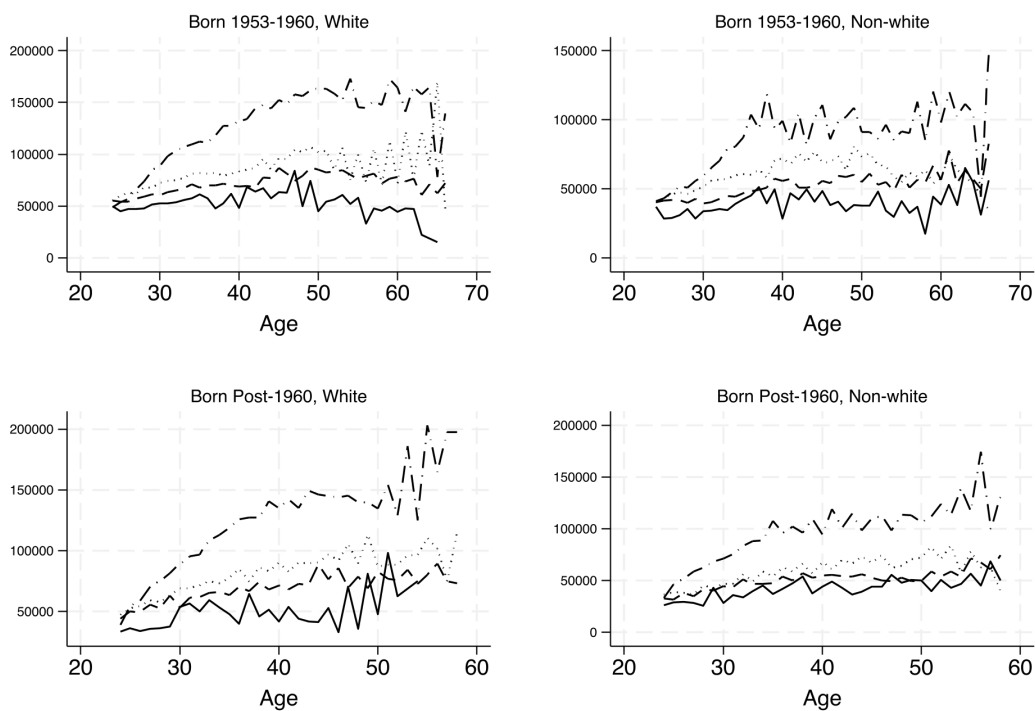
Table 1: Summary Statistics of the Risk Measures

	Mean	SD	Bot 10% Mean	Top 10% Mean	Person-Years	Persons
Gamma (Hourly)	0.0160	0.1119	-0.1606	0.2479	82,112	9,340
Alpha (Hourly)	0.0902	0.2761	-0.1193	0.7388	101,315	10,621
Gamma (Annual)	0.0248	0.1630	-0.2218	0.3618	80,661	9,281
Alpha (Annual)	0.1286	0.4815	-0.1988	1.2015	99,639	10,546

Table 2: Stratified Means of Gamma and Alpha by Demographic Group

Group	Category	Gamma (hourly)	Alpha (hourly)	Gamma (annual)	Alpha (annual)	N
Hourly Wage Quintile	1st Quintile	0.0191	0.1262	0.0313	0.2059	14,932
Hourly Wage Quintile	2nd Quintile	0.0135	0.0856	0.0236	0.1249	16,044
Hourly Wage Quintile	3rd Quintile	0.0148	0.0722	0.0209	0.0957	16,381
Hourly Wage Quintile	4th Quintile	0.0146	0.0636	0.0198	0.0785	16,601
Hourly Wage Quintile	5th Quintile	0.0181	0.0842	0.0247	0.0915	16,329
Annual Earnings Quintile	1st Quintile	0.0183	0.1521	0.0390	0.2709	14,363
Annual Earnings Quintile	2nd Quintile	0.0142	0.0837	0.0217	0.1223	16,476
Annual Earnings Quintile	3rd Quintile	0.0140	0.0709	0.0201	0.0935	17,058
Annual Earnings Quintile	4th Quintile	0.0146	0.0614	0.0196	0.0701	17,095
Annual Earnings Quintile	5th Quintile	0.0191	0.0854	0.0251	0.0902	17,067
Tenure Bin	0-1	0.0165	0.1078	0.0267	0.1698	39,980
Tenure Bin	2-5	0.0163	0.0751	0.0242	0.0906	18,839
Tenure Bin	6+	0.0148	0.0697	0.0217	0.0823	23,293
Age Bin	22-29	0.0170	0.0929	0.0269	0.1571	16,303
Age Bin	30-37	0.0166	0.0861	0.0252	0.1241	24,279
Age Bin	38-45	0.0145	0.0867	0.0223	0.1174	18,541
Age Bin	46-53	0.0152	0.0870	0.0218	0.1124	13,749
Age Bin	54-61	0.0154	0.0907	0.0274	0.1221	7,988
Age Bin	62-69	0.0237	0.1585	0.0482	0.2073	1,252
Cohort	born pre-1944	0.0163	0.0871	0.0252	0.1112	19,878
Cohort	born 1944-1952	0.0148	0.0865	0.0224	0.1224	21,560
Cohort	born 1953-1960	0.0157	0.0911	0.0250	0.1415	19,570
Cohort	born post-1960	0.0171	0.0959	0.0264	0.1381	21,104
Race	White	0.0161	0.0859	0.0242	0.1130	54,361
Race	Black	0.0146	0.1011	0.0256	0.1677	20,207
Race	Native American	0.0197	0.0901	0.0282	0.1405	1,834
Race	Asian or Pacific Islander	0.0180	0.0831	0.0277	0.1047	752
Race	Latino	0.0183	0.0891	0.0267	0.1275	4,490
Race	Other or Unknown	0.0185	0.1085	0.0196	0.1320	468
Education	Less than HS	0.0150	0.1000	0.0263	0.1660	13,247
Education	High School Graduate	0.0148	0.0921	0.0240	0.1376	34,816
Education	Some College	0.0176	0.0888	0.0272	0.1123	11,987
Education	Bachelor's Degree	0.0182	0.0797	0.0239	0.0971	14,697
Education	Bachelors+	0.0165	0.0832	0.0236	0.0981	6,764

Figure 1: Average Annual Earnings by Age, Born 1953-1960 & Post-1960



Solid = HS Dropout; Dashed = HS Graduate; Dotted = Some College; Dash-dot = College Graduate.
Annual earnings in 2024 dollars.

Figure 2: Distribution of Gamma and Alpha

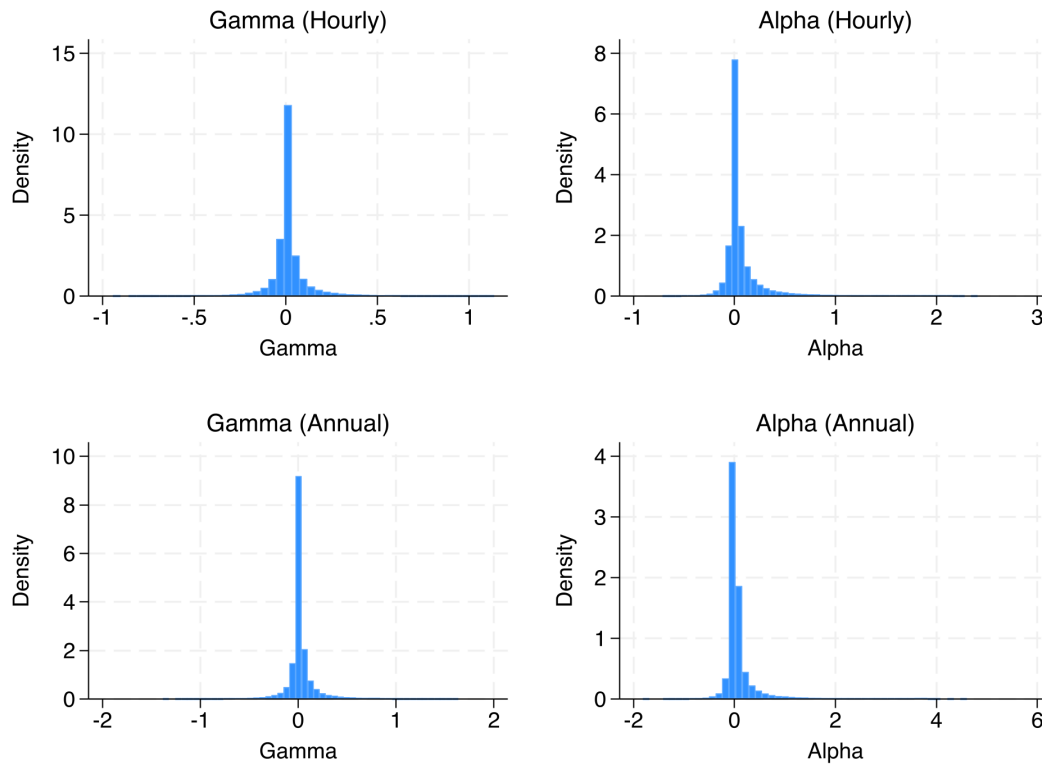


Figure 3: Distribution of Gamma and Alpha Across Age

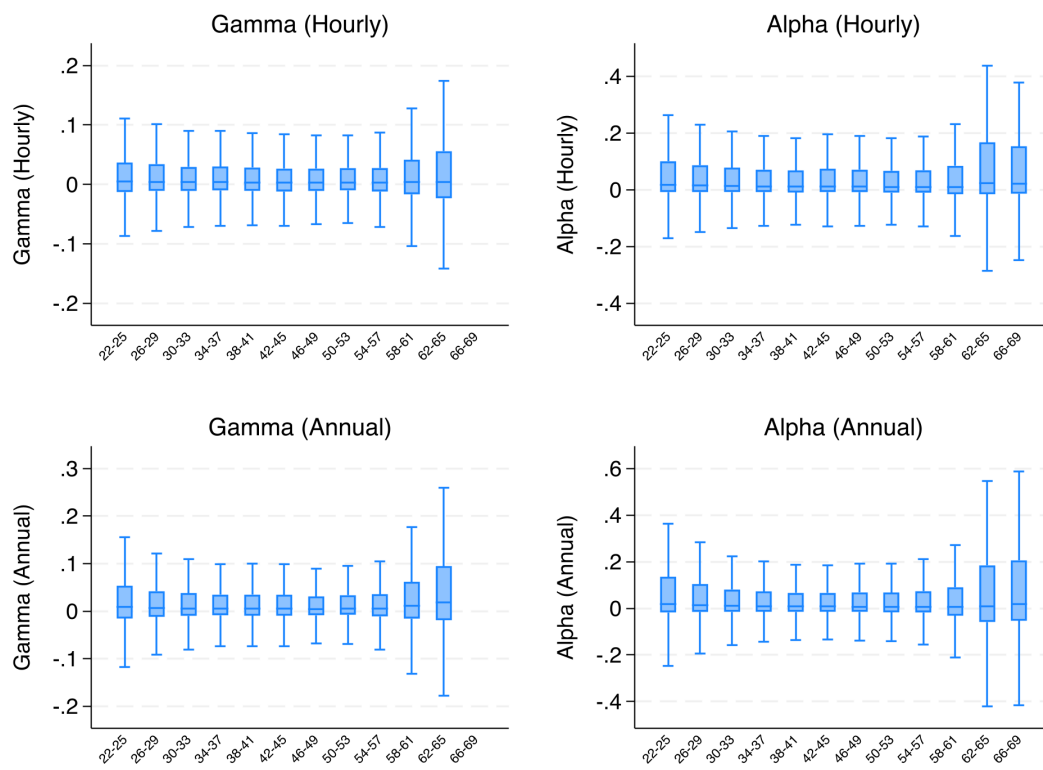


Table 3: Gamma and Alpha Regressions: OLS Results (Coefficients x 100)

	Gamma, Hourly Wage		Gamma, Annual Earnings		Alpha, Hourly Wage		Alpha, Annual Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No Occ/Ind	All Controls	No Occ/Ind	All Controls	No Occ/Ind	All Controls	No Occ/Ind	All Controls
High School Graduate	0.0682 (0.121)	0.119 (0.124)	0.123 (0.179)	0.222 (0.182)	0.564** (0.262)	1.134*** (0.264)	0.696 (0.457)	1.377*** (0.462)
Some College	0.367** (0.152)	0.394** (0.158)	0.609*** (0.224)	0.619*** (0.233)	1.013*** (0.332)	1.415*** (0.342)	0.680 (0.579)	1.089* (0.597)
Bachelor's Degree	0.454*** (0.152)	0.469*** (0.166)	0.528** (0.224)	0.476* (0.244)	1.615*** (0.333)	1.603*** (0.359)	3.227*** (0.580)	3.164*** (0.627)
Bachelors+	0.290 (0.188)	0.276 (0.211)	0.648** (0.278)	0.587* (0.312)	2.086*** (0.411)	1.937*** (0.458)	4.509*** (0.719)	4.326*** (0.803)
22-29	0.00556*** (0.00200)	0.00556*** (0.00200)	0.00345 (0.00294)	0.00290 (0.00295)	-0.0132*** (0.00438)	-0.0130*** (0.00436)	-0.0150* (0.00764)	-0.0188** (0.00763)
30-37	0.00427*** (0.00157)	0.00434*** (0.00158)	0.00405* (0.00231)	0.00404* (0.00232)	-0.00549 (0.00346)	-0.00478 (0.00344)	-0.00656 (0.00602)	-0.00636 (0.00600)
38-45	0.000816 (0.00136)	0.000916 (0.00136)	0.00177 (0.00199)	0.00186 (0.00199)	-0.000477 (0.00299)	-0.000243 (0.00298)	0.00412 (0.00520)	0.00424 (0.00518)
54-61	-0.000696 (0.00162)	-0.00112 (0.00162)	0.00443* (0.00250)	0.00335 (0.00250)	0.00176 (0.00347)	-0.000208 (0.00345)	-0.00106 (0.00621)	-0.00466 (0.00619)
62-69	0.00687** (0.00339)	0.00537 (0.00340)	0.0238*** (0.00587)	0.0218*** (0.00588)	0.0478*** (0.00585)	0.0413*** (0.00583)	0.0445*** (0.0117)	0.0362*** (0.0117)
Probability of Recession	2.121 (4.872)	4.047 (4.930)	-4.437 (7.267)	2.837 (7.340)	0.583 (0.922)	2.858*** (0.926)	3.287** (1.629)	8.034*** (1.637)
5-year moving average of AEP	0.00328 (0.00221)	0.00264 (0.00242)	0.0267*** (0.00322)	0.0236*** (0.00354)	0.136*** (0.00474)	0.120*** (0.00518)	0.381*** (0.00822)	0.367*** (0.00900)
Out of Labor Force	-0.525 (0.370)	-0.675* (0.375)	-1.344** (0.597)	-2.027*** (0.603)	7.685*** (0.727)	5.730*** (0.733)	16.76*** (1.418)	11.26*** (1.428)
2-5	-0.000399 (0.00112)	0.000246 (0.00113)	-0.00229 (0.00164)	-0.000273 (0.00166)	-0.0204*** (0.00254)	-0.0142*** (0.00255)	-0.0537*** (0.00440)	-0.0357*** (0.00443)
6+	-0.00127 (0.00110)	-0.000797 (0.00113)	-0.00322** (0.00162)	-0.00146 (0.00166)	-0.0191*** (0.00248)	-0.0148*** (0.00252)	-0.0436*** (0.00432)	-0.0266*** (0.00440)
Census Division FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Race FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation FE	No	Yes	No	Yes	No	Yes	No	Yes
Industry FE	No	Yes	No	Yes	No	Yes	No	Yes
R-squared	0.001	0.004	0.003	0.006	0.022	0.037	0.040	0.052
N	82112	82112	80661	80661	101315	101315	99639	99639

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Gamma Regressions: F-Test Results

	Hourly Wage		Annual Earnings	
	No Occ/Ind	All Controls	No Occ/Ind	All Controls
	(1)	(2)	(3)	(4)
Census Division FE	2.21** (0.0188)	1.87* (0.0518)	2.66*** (0.0044)	2.47*** (0.0081)
Year FE	1.11 (0.3117)	1.11 (0.3022)	1.63** (0.0137)	2.30*** (0.0000)
Race FE	1.43 (0.2112)	0.97 (0.4324)	0.38 (0.8615)	0.46 (0.8058)
Cohort FE	4.46*** (0.0039)	4.68*** (0.0028)	3.65** (0.0121)	4.12*** (0.0062)
Occupation FE	—	1.96*** (0.0000)	—	1.92*** (0.0000)
Industry FE	—	1.93*** (0.0012)	—	2.27*** (0.0001)
Education FE	3.90*** (0.0036)	3.01** (0.0170)	3.57*** (0.0065)	2.13* (0.0738)
Age Bin FE	3.51*** (0.0036)	3.19*** (0.0070)	4.15*** (0.0009)	3.44*** (0.0041)
Tenure FE	0.68 (0.5083)	0.44 (0.6458)	2.13 (0.1186)	0.42 (0.6564)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 5: Alpha Regressions: F-Test Results

	Hourly Wage		Annual Earnings	
	No Occ/Ind	All Controls	No Occ/Ind	All Controls
	(1)	(2)	(3)	(4)
Census Division FE	14.40*** (0.0000)	7.55*** (0.0000)	12.30*** (0.0000)	9.27*** (0.0000)
Year FE	8.85*** (0.0000)	11.03*** (0.0000)	4.97*** (0.0000)	10.18*** (0.0000)
Race FE	4.44*** (0.0005)	7.49*** (0.0000)	5.80*** (0.0000)	9.43*** (0.0000)
Cohort FE	2.63** (0.0484)	4.05*** (0.0069)	2.50* (0.0572)	3.50** (0.0148)
Occupation FE	—	7.05*** (0.0000)	—	4.78*** (0.0000)
Industry FE	—	10.12*** (0.0000)	—	6.93*** (0.0000)
Education FE	9.48*** (0.0000)	6.79*** (0.0000)	17.10*** (0.0000)	9.73*** (0.0000)
Age Bin FE	15.89*** (0.0000)	12.96*** (0.0000)	4.67*** (0.0003)	5.00*** (0.0001)
Tenure FE	42.69*** (0.0000)	22.38*** (0.0000)	88.12*** (0.0000)	36.00*** (0.0000)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 6: Gamma and Alpha Regressions: Stepwise Selection Results (Coefficients x 100)

	Gamma, Hourly Wage		Gamma, Annual Earnings		Alpha, Hourly Wage		Alpha, Annual Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No Occ/Ind	All Controls	No Occ/Ind	All Controls	No Occ/Ind	All Controls	No Occ/Ind	All Controls
Probability of Recession			0.922*	0.873*	1.905*	1.966*	3.836*	3.975**
			(0.432)	(0.432)	(0.850)	(0.844)	(1.496)	(1.488)
5-year moving average of AEP			0.0275***	0.0220***	0.136***	0.120***	0.380***	0.367***
			(0.00306)	(0.00326)	(0.00474)	(0.00518)	(0.00821)	(0.00900)
Out of Labor Force			-1.221*	-1.951**	7.685***	5.730***	16.77***	11.26***
			(0.594)	(0.601)	(0.727)	(0.733)	(1.418)	(1.428)
Constant	0.0134***	0.0188**	-0.00624	0.00693	0.0329*	0.0415*	-0.0390	-0.0650
	(0.00163)	(0.00605)	(0.00571)	(0.00998)	(0.0145)	(0.0192)	(0.0250)	(0.0334)
Census Division FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE			✓	✓	✓	✓	✓	✓
Race FE					✓	✓	✓	✓
Cohort FE			✓	✓	✓	✓		✓
Occupation FE		✓		✓		✓		✓
Industry FE		✓		✓		✓		✓
Education FE	✓	✓	✓		✓	✓	✓	✓
Age Bin FE	✓		✓	✓	✓	✓	✓	✓
Tenure FE					✓	✓	✓	✓
Census Division FE Available	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE Available	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Race FE Available	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE Available	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation FE Available	No	Yes	No	Yes	No	Yes	No	Yes
Industry FE Available	No	Yes	No	Yes	No	Yes	No	Yes
Education FE Available	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age Bin FE Available	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Tenure FE Available	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.001	0.003	0.002	0.006	0.022	0.037	0.040	0.052
N	82112	82112	80661	80661	101315	101315	99639	99639

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7: Correlation of ML Predictions, Hourly Earnings

	Gamma			Alpha		
	NN	RF	LASSO	NN	RF	LASSO
NN	1.000			1.000		
RF	0.283	1.000		0.603	1.000	
LASSO	0.724	0.387	1.000	0.934	0.657	1.000

Table 8: Correlation of ML Predictions, Annual Earnings

	Gamma			Alpha		
	NN	RF	LASSO	NN	RF	LASSO
NN	1.000			1.000		
RF	0.306	1.000		0.697	1.000	
LASSO	0.737	0.388	1.000	0.837	0.640	1.000

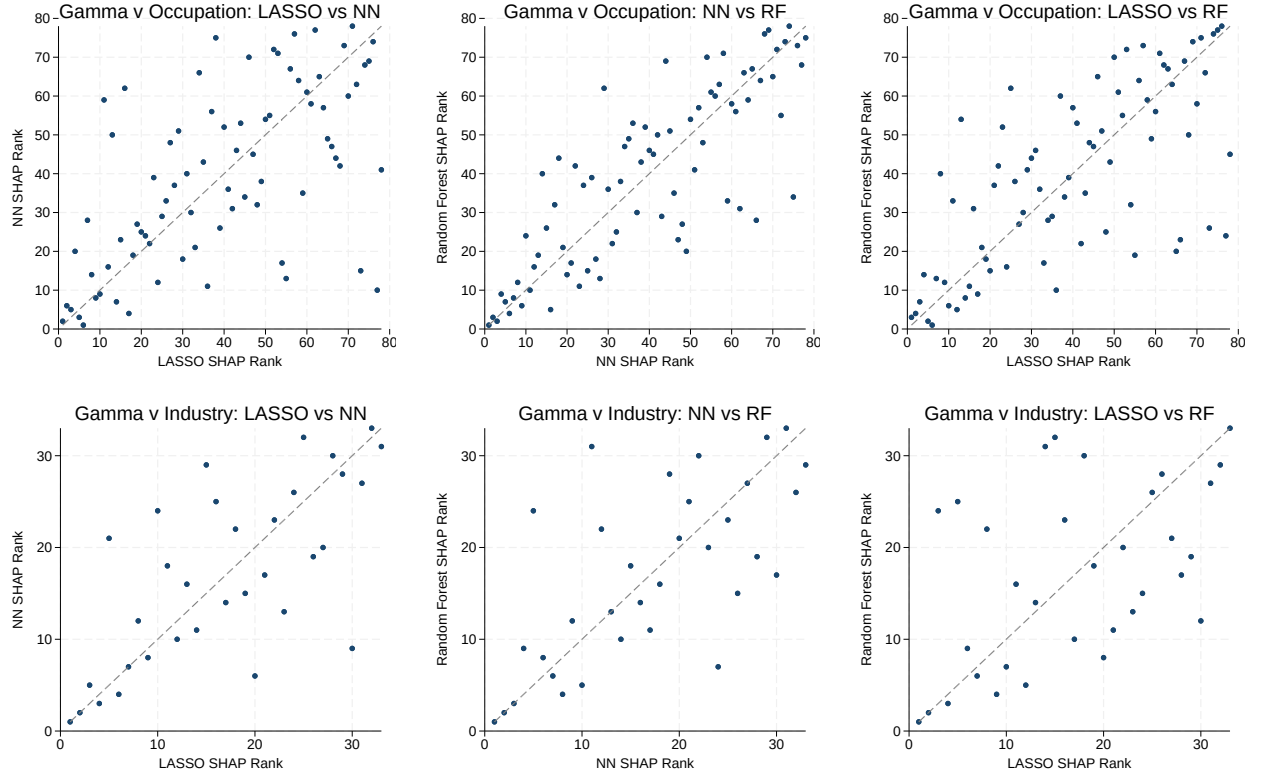


Figure 4: Gamma SHAP Rank Comparisons Across Methods, Annual Earnings (fearn). Top row: occupation; bottom row: industry.

Table 9: Out-of-Sample Test MSE by Method

	Gamma		Alpha	
	Hourly	Annual	Hourly	Annual
NN	0.01250	0.02575	0.07880	0.23037
Random Forest	0.01254	0.02584	0.07893	0.23176
LASSO	0.01250	0.02575	0.07850	0.23073
Predict Train Mean	0.01250	0.02574	0.08041	0.23912
N (test)	12,227	12,009	15,082	14,832

Table 10: Group SHAP Shares of Controls, Alpha (Average —SHAP— per Variable x 100)

	Hourly Wage			Annual Earnings		
	NN	RF	LASSO	NN	RF	LASSO
Census Division	0.0301	0.0247	0.0464	0.0205	0.0248	0.0255
Year	0.0173	0.0157	0.0300	0.0143	0.0131	0.0253
Race	0.0308	0.0257	0.0411	0.0364	0.0503	0.0453
Cohort	0.0215	0.0289	0.0305	0.0426	0.0353	0.0141
Occupation	0.0083	0.0077	0.0112	0.0077	0.0065	0.0093
Industry	0.0236	0.0195	0.0373	0.0226	0.0248	0.0309
Education	0.0151	0.0170	0.0207	0.0760	0.0219	0.1156
Age	0.0301	0.0343	0.0426	0.0407	0.0302	0.0263
Tenure	0.0461	0.5417	0.1022	0.0917	0.4900	0.1717

Table 11: Mean Predicted Gamma and Alpha by Demographic Group, Hourly Earnings

Group	Category	Actual	Gamma				Alpha				
			NN	RF	LASSO	N	Actual	NN	RF	LASSO	N
Education	Less than HS	0.0150	0.0186	0.0147	0.0143	13,247	0.1000	0.0939	0.0949	0.0984	17,294
Education	High School Graduate	0.0148	0.0186	0.0148	0.0145	34,816	0.0921	0.0888	0.0900	0.0902	43,010
Education	Some College	0.0176	0.0186	0.0161	0.0173	11,987	0.0888	0.0862	0.0884	0.0876	14,528
Education	Bachelor's Degree	0.0182	0.0186	0.0169	0.0178	14,697	0.0797	0.0824	0.0817	0.0792	17,489
Education	Bachelors+	0.0165	0.0186	0.0166	0.0163	6,764	0.0832	0.0851	0.0831	0.0822	8,220
Race	White	0.0162	0.0186	0.0157	0.0157	54,090	0.0859	0.0850	0.0858	0.0850	65,542
Race	Black	0.0145	0.0186	0.0143	0.0140	20,065	0.1011	0.0946	0.0941	0.0982	25,891
Race	Native American	0.0197	0.0186	0.0168	0.0194	1,820	0.0896	0.0884	0.0938	0.0921	2,193
Race	Asian or Pacific Islander	0.0176	0.0187	0.0186	0.0200	697	0.0779	0.0791	0.0903	0.0721	857
Race	Latino	0.0184	0.0186	0.0172	0.0191	4,371	0.0888	0.0900	0.0913	0.0871	5,478
Race	Other or Unknown	0.0185	0.0186	0.0193	0.0196	468	0.1087	0.0989	0.1074	0.1013	580
Cohort	born pre-1944	0.0163	0.0186	0.0155	0.0160	19,850	0.0871	0.0859	0.0880	0.0865	24,778
Cohort	born 1944-1952	0.0149	0.0186	0.0143	0.0135	21,430	0.0866	0.0869	0.0866	0.0859	25,694
Cohort	born 1953-1960	0.0157	0.0186	0.0154	0.0154	19,459	0.0909	0.0893	0.0882	0.0884	23,984
Cohort	born post-1960	0.0172	0.0186	0.0169	0.0177	20,772	0.0958	0.0894	0.0914	0.0936	26,085
Age Bin	22-29	0.0171	0.0186	0.0164	0.0166	16,169	0.0927	0.0896	0.0915	0.0882	19,269
Age Bin	30-37	0.0166	0.0186	0.0160	0.0163	24,113	0.0859	0.0859	0.0866	0.0863	28,867
Age Bin	38-45	0.0145	0.0186	0.0143	0.0138	18,415	0.0867	0.0858	0.0849	0.0842	22,310
Age Bin	46-53	0.0153	0.0186	0.0142	0.0154	13,654	0.0872	0.0878	0.0854	0.0879	16,605
Age Bin	54-61	0.0155	0.0186	0.0160	0.0150	7,922	0.0908	0.0880	0.0881	0.0900	10,724
Age Bin	62-69	0.0244	0.0187	0.0242	0.0242	1,238	0.1587	0.1136	0.1397	0.1514	2,766
Tenure Bin	0-1	0.0165	0.0186	0.0154	0.0155	39,662	0.1078	0.0943	0.1049	0.0999	50,885
Tenure Bin	2-5	0.0163	0.0186	0.0155	0.0158	18,671	0.0750	0.0841	0.0739	0.0815	21,501
Tenure Bin	6+	0.0149	0.0186	0.0158	0.0158	23,178	0.0697	0.0792	0.0702	0.0738	28,155

Table 12: Mean Predicted Gamma and Alpha by Demographic Group, Annual Earnings

Group	Category	Gamma					Alpha				
		Actual	NN	RF	LASSO	N	Actual	NN	RF	LASSO	N
Education	Less than HS	0.0263	0.0276	0.0260	0.0270	12,793	0.1660	0.1690	0.1492	0.1654	16,727
Education	High School Graduate	0.0240	0.0278	0.0245	0.0242	34,355	0.1376	0.1527	0.1376	0.1409	42,477
Education	Some College	0.0272	0.0282	0.0267	0.0285	11,816	0.1123	0.1307	0.1225	0.1152	14,362
Education	Bachelor's Degree	0.0239	0.0278	0.0248	0.0243	14,515	0.0971	0.1113	0.1020	0.0917	17,278
Education	Bachelors+	0.0236	0.0277	0.0249	0.0238	6,581	0.0981	0.1138	0.1063	0.0980	8,031
Race	White	0.0242	0.0278	0.0246	0.0245	52,885	0.1129	0.1280	0.1136	0.1141	64,168
Race	Black	0.0256	0.0280	0.0262	0.0273	19,929	0.1676	0.1759	0.1628	0.1667	25,709
Race	Native American	0.0280	0.0276	0.0282	0.0289	1,778	0.1394	0.1494	0.1424	0.1390	2,156
Race	Asian or Pacific Islander	0.0294	0.0273	0.0264	0.0252	679	0.1030	0.1209	0.1222	0.1028	842
Race	Latino	0.0262	0.0276	0.0267	0.0261	4,321	0.1260	0.1448	0.1382	0.1316	5,422
Race	Other or Unknown	0.0196	0.0274	0.0201	0.0141	468	0.1321	0.1400	0.1485	0.1296	578
Cohort	born pre-1944	0.0252	0.0274	0.0249	0.0252	18,365	0.1113	0.1289	0.1168	0.1156	23,055
Cohort	born 1944-1952	0.0224	0.0279	0.0245	0.0235	21,252	0.1221	0.1399	0.1239	0.1223	25,494
Cohort	born 1953-1960	0.0250	0.0278	0.0257	0.0260	19,521	0.1410	0.1533	0.1401	0.1411	24,041
Cohort	born post-1960	0.0266	0.0282	0.0255	0.0266	20,922	0.1383	0.1448	0.1331	0.1371	26,285
Age Bin	22-29	0.0272	0.0281	0.0284	0.0277	16,476	0.1572	0.1634	0.1580	0.1583	19,603
Age Bin	30-37	0.0251	0.0282	0.0254	0.0261	24,324	0.1240	0.1409	0.1269	0.1244	29,060
Age Bin	38-45	0.0223	0.0277	0.0225	0.0228	18,437	0.1171	0.1358	0.1168	0.1194	22,347
Age Bin	46-53	0.0219	0.0272	0.0218	0.0222	13,306	0.1122	0.1264	0.1103	0.1106	16,316
Age Bin	54-61	0.0275	0.0275	0.0274	0.0271	6,671	0.1216	0.1379	0.1189	0.1246	9,622
Age Bin	62-69	0.0492	0.0283	0.0493	0.0448	846	0.2073	0.1606	0.1951	0.2027	1,927
Tenure Bin	0-1	0.0267	0.0280	0.0275	0.0268	39,275	0.1697	0.1643	0.1691	0.1566	50,330
Tenure Bin	2-5	0.0244	0.0278	0.0246	0.0250	18,426	0.0904	0.1265	0.0879	0.1135	21,250
Tenure Bin	6+	0.0218	0.0276	0.0216	0.0229	22,359	0.0822	0.1124	0.0857	0.0910	27,295

Table 13: Mean Predicted Alpha by Education and Age Bin, Hourly Earnings

Actual							Neural Network						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.1234	0.1008	0.0905	0.0847	0.0881	0.1710	Less than HS	0.0967	0.0929	0.0911	0.0907	0.0923	0.1229
High School Graduate	0.0936	0.0933	0.0907	0.0875	0.0802	0.1566	High School Graduate	0.0900	0.0875	0.0875	0.0886	0.0879	0.1164
Some College	0.0851	0.0806	0.0864	0.0896	0.1144	0.1554	Some College	0.0884	0.0841	0.0839	0.0865	0.0877	0.1109
Bachelor's Degree	0.0703	0.0694	0.0800	0.0887	0.0873	0.1728	Bachelor's Degree	0.0828	0.0805	0.0813	0.0837	0.0829	0.1019
Bachelors+	0.0563	0.0684	0.0781	0.0855	0.1077	0.1282	Bachelors+	0.0845	0.0818	0.0821	0.0871	0.0860	0.1074

Random Forest							LASSO						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.1024	0.0962	0.0879	0.0871	0.0889	0.1558	Less than HS	0.0994	0.0976	0.0919	0.0919	0.0962	0.1720
High School Graduate	0.0917	0.0896	0.0886	0.0853	0.0865	0.1374	High School Graduate	0.0889	0.0890	0.0875	0.0896	0.0906	0.1546
Some College	0.0908	0.0843	0.0830	0.0883	0.0966	0.1477	Some College	0.0879	0.0853	0.0833	0.0878	0.0911	0.1480
Bachelor's Degree	0.0817	0.0781	0.0790	0.0829	0.0836	0.1330	Bachelor's Degree	0.0755	0.0768	0.0767	0.0813	0.0816	0.1297
Bachelors+	0.0822	0.0776	0.0788	0.0833	0.0884	0.1183	Bachelors+	0.0794	0.0766	0.0747	0.0843	0.0851	0.1361

Table 14: Mean Predicted Alpha by Education and Age Bin, Annual Earnings

Actual						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.2432	0.1660	0.1470	0.1306	0.1388	0.2281
High School Graduate	0.1662	0.1403	0.1257	0.1126	0.1065	0.1839
Some College	0.1182	0.1062	0.1080	0.1036	0.1232	0.2367
Bachelor's Degree	0.0826	0.0862	0.0993	0.1034	0.1171	0.2112
Bachelors+	0.0764	0.0837	0.0835	0.0977	0.1323	0.1963

Neural Network						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.1984	0.1760	0.1633	0.1466	0.1599	0.1855
High School Graduate	0.1711	0.1532	0.1476	0.1313	0.1430	0.1684
Some College	0.1451	0.1287	0.1265	0.1182	0.1322	0.1544
Bachelor's Degree	0.1200	0.1082	0.1093	0.1069	0.1136	0.1377
Bachelors+	0.1308	0.1123	0.1085	0.1092	0.1173	0.1481

Random Forest						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.1972	0.1544	0.1304	0.1238	0.1330	0.2265
High School Graduate	0.1643	0.1364	0.1272	0.1133	0.1205	0.1999
Some College	0.1405	0.1196	0.1117	0.1095	0.1250	0.2088
Bachelor's Degree	0.1135	0.0991	0.0978	0.0944	0.1009	0.1674
Bachelors+	0.1329	0.1067	0.0975	0.0994	0.1041	0.1707

LASSO						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.2001	0.1655	0.1553	0.1402	0.1572	0.2660
High School Graduate	0.1648	0.1370	0.1320	0.1195	0.1318	0.2160
Some College	0.1389	0.1109	0.1074	0.0967	0.1132	0.1914
Bachelor's Degree	0.1133	0.0876	0.0875	0.0776	0.0898	0.1544
Bachelors+	0.1268	0.0970	0.0902	0.0892	0.0969	0.1679

Table 15: Characteristics of Top and Bottom Risk Deciles by Age, Gamma (Sorted by Actual Risk; Percent Shares)

Panel A: Hourly Wage

		Ages 22-33			Ages 34-57			Ages 58-69		
		All	Bottom 10%	Top 10%	All	Bottom 10%	Top 10%	All	Bottom 10%	Top 10%
Education	Less than HS	13.6	16.0	16.0	18.2	23.3	18.1	22.3	19.0	17.1
Education	High School Graduate	47.4	48.5	42.5	40.1	40.2	42.2	33.1	35.9	35.4
Education	Some College	16.2	16.5	16.9	14.9	15.2	15.1	13.3	16.7	15.6
Education	Bachelor's Degree	17.4	13.5	17.7	17.0	12.3	17.2	17.3	12.7	13.3
Education	Bachelors+	5.4	5.5	6.8	9.7	9.0	7.4	14.0	15.7	18.5
Race	White	60.7	53.4	56.5	61.4	54.8	56.7	69.8	73.5	72.9
Race	Black	29.6	35.5	31.4	26.8	34.0	30.3	20.0	14.8	16.1
Race	Native American	2.0	2.0	2.9	2.2	1.6	2.7	2.5	3.2	2.6
Race	Asian or Pacific Islander	0.7	0.8	1.1	1.3	1.2	1.3	1.9	1.3	1.3
Race	Latino	6.6	7.8	7.5	7.5	7.4	8.1	4.7	5.8	5.8
Race	Other or Unknown	0.4	0.5	0.6	0.8	1.0	0.9	1.1	1.3	1.3
Cohort	born pre-1944	5.0	5.1	4.4	25.1	20.3	23.6	54.5	60.0	59.4
Cohort	born 1944-1952	19.6	15.2	14.7	19.6	17.8	15.8	26.0	22.6	23.9
Cohort	born 1953-1960	22.3	17.3	19.3	20.2	21.2	19.3	19.5	17.4	16.8
Cohort	born post-1960	53.1	62.5	61.6	35.1	40.6	41.4	0.0	0.0	0.0
Tenure Bin	0-1	59.6	67.6	66.5	45.0	57.1	51.7	34.8	44.2	41.7
Tenure Bin	2-5	27.5	23.9	24.4	21.8	20.1	22.3	19.1	15.6	14.4
Tenure Bin	6+	12.9	8.5	9.1	33.2	22.8	26.0	46.1	40.3	43.9
Census Division	Unknown Census division	0.5	0.5	0.5	0.5	0.5	1.4	0.3	1.0	0.2
Census Division	New England	3.4	2.0	2.1	3.8	3.2	2.8	5.2	3.2	5.5
Census Division	Middle Atlantic	11.2	11.1	10.0	11.8	10.1	9.3	12.8	8.4	9.7
Census Division	East North Central	15.5	15.1	16.0	15.5	13.5	14.7	15.0	21.6	13.7
Census Division	West North Central	9.4	8.8	10.1	8.7	10.2	9.1	9.0	9.0	11.3
Census Division	South Atlantic	23.6	24.2	20.5	23.1	27.1	25.0	20.7	20.3	21.9
Census Division	East South Central	8.4	7.3	10.4	7.8	6.9	7.5	6.8	5.8	5.8
Census Division	West South Central	10.9	13.2	12.0	10.2	10.1	12.4	10.2	9.4	10.1
Census Division	Mountain	5.0	4.0	5.5	5.0	5.3	4.0	6.0	4.8	9.4
Census Division	Pacific	12.2	13.7	12.8	13.6	13.1	13.6	14.0	16.5	12.5

Panel B: Annual Earnings

		Ages 22-33			Ages 34-57			Ages 58-69		
		All	Bottom 10%	Top 10%	All	Bottom 10%	Top 10%	All	Bottom 10%	Top 10%
Education	Less than HS	13.8	21.1	21.4	17.9	22.9	20.9	19.5	21.8	10.7
Education	High School Graduate	47.6	47.9	43.9	40.2	43.5	39.9	32.1	28.2	33.9
Education	Some College	16.1	15.2	16.3	15.0	13.8	15.9	13.6	13.2	15.0
Education	Bachelor's Degree	17.3	12.1	12.7	17.2	11.2	15.3	18.9	19.5	17.1
Education	Bachelors+	5.3	3.6	5.7	9.7	8.6	8.2	15.9	17.3	23.3
Race	White	60.4	45.7	52.1	61.3	52.5	54.9	69.9	68.0	73.6
Race	Black	30.0	44.5	36.2	26.8	35.4	32.0	19.1	20.0	16.0
Race	Native American	2.0	1.7	1.8	2.2	1.6	3.1	2.9	4.8	3.2
Race	Asian or Pacific Islander	0.7	1.1	0.8	1.3	0.9	1.3	2.0	0.8	2.4
Race	Latino	6.6	6.4	9.0	7.6	8.6	7.8	4.8	6.4	4.0
Race	Other or Unknown	0.4	0.6	0.2	0.8	1.0	0.9	1.3	0.0	0.8
Cohort	born pre-1944	4.9	5.3	4.4	24.6	22.2	24.3	50.8	56.0	52.0
Cohort	born 1944-1952	19.5	17.3	14.4	19.7	17.5	18.3	29.0	28.8	30.4
Cohort	born 1953-1960	22.5	19.8	23.4	20.4	21.9	19.4	20.1	15.2	17.6
Cohort	born post-1960	53.1	57.6	57.8	35.3	38.5	38.0	0.0	0.0	0.0
Tenure Bin	0-1	59.9	73.5	70.7	45.2	58.0	57.1	33.7	46.7	41.0
Tenure Bin	2-5	27.4	21.1	20.2	21.8	19.3	21.1	20.6	19.6	19.8
Tenure Bin	6+	12.7	5.4	9.1	33.0	22.7	21.8	45.8	33.7	39.2
Census Division	Unknown Census division	0.5	0.5	0.6	0.5	0.9	1.4	0.4	0.5	0.8
Census Division	New England	3.3	2.4	2.5	3.8	3.2	1.9	5.5	8.0	3.2
Census Division	Middle Atlantic	11.3	11.1	9.1	11.8	11.4	9.7	11.8	7.2	17.6
Census Division	East North Central	15.6	15.1	18.1	15.4	12.5	17.3	14.5	15.6	17.2
Census Division	West North Central	9.4	8.0	7.6	8.8	8.9	8.9	9.1	9.6	9.9
Census Division	South Atlantic	23.5	23.9	21.2	23.2	25.6	23.4	19.9	21.1	19.5
Census Division	East South Central	8.4	8.5	9.7	7.8	8.7	7.1	7.7	6.4	4.8
Census Division	West South Central	10.9	13.4	13.1	10.0	11.2	12.6	10.5	14.4	8.0
Census Division	Mountain	4.9	4.1	4.6	5.0	4.4	4.2	6.5	4.0	9.5
Census Division	Pacific	12.2	12.8	13.3	13.6	13.3	13.5	14.1	13.2	9.6

Table 16: Characteristics of Top and Bottom Risk Deciles by Age, Alpha (Sorted by Ensemble-Predicted and Actual Risk; Percent Shares)

Panel A: Hourly Wage

		Ages 22-33					Ages 34-57					Ages 58-69				
		All	Bottom 10%	Top 10%			All	Bottom 10%	Top 10%			All	Bottom 10%	Top 10%		
		Pred	Act	Pred	Act		Pred	Act	Pred	Act		Pred	Act	Pred	Act	
Education	Less than HS	14.5	9.5	13.8	20.5	21.1	18.7	14.7	14.1	29.2	17.4	24.0	14.8	17.0	30.7	19.9
Education	High School Graduate	48.2	43.2	44.5	54.9	51.3	40.5	32.1	38.8	47.3	47.2	33.2	29.3	34.2	34.4	34.0
Education	Some College	15.7	15.8	15.2	15.2	12.2	14.9	15.5	16.9	12.0	13.9	13.2	16.1	12.2	13.5	13.8
Education	Bachelor's Degree	16.5	24.9	19.9	6.8	11.3	16.5	25.2	20.5	7.5	14.1	16.8	24.3	20.8	13.0	19.1
Education	Bachelors+	5.1	6.6	6.6	2.7	4.1	9.4	12.6	9.7	3.9	7.5	12.8	15.5	15.8	8.4	13.2
Race	White	58.9	78.4	60.1	40.9	49.5	60.5	81.1	61.7	46.0	54.7	69.6	84.5	76.4	74.5	72.7
Race	Black	31.2	14.5	28.4	51.6	40.2	28.2	10.5	26.7	44.5	34.4	20.4	9.1	15.0	18.6	16.4
Race	Native American	1.9	1.4	1.6	2.2	1.4	2.0	0.9	2.3	2.1	2.1	2.1	0.5	2.7	1.8	2.7
Race	Asian or Pacific Islander	0.7	0.7	1.1	0.7	0.7	1.3	1.7	1.0	0.5	1.2	1.6	2.7	0.0	0.5	1.8
Race	Latino	6.8	5.1	7.9	4.1	8.1	7.3	5.7	8.1	6.1	6.6	5.3	3.2	5.5	2.7	5.5
Race	Other or Unknown	0.4	0.0	0.8	0.5	0.3	0.8	0.0	0.1	0.8	1.0	0.9	0.0	0.5	1.8	0.9
Cohort	born pre-1944	4.7	12.4	5.9	1.8	3.1	24.3	38.5	20.2	20.1	16.4	53.3	31.8	52.7	75.0	62.7
Cohort	born 1944-1952	19.1	34.2	14.1	11.5	13.7	18.8	7.0	15.9	19.3	18.4	24.8	11.8	23.2	21.8	24.5
Cohort	born 1953-1960	22.4	24.3	19.3	17.6	17.2	20.6	3.3	21.6	29.2	25.5	20.2	54.1	20.9	3.2	11.8
Cohort	born post-1960	53.9	29.1	60.8	69.2	66.0	36.3	51.2	42.3	31.5	39.7	1.7	2.3	3.2	0.0	0.9
Tenure Bin	0-1	60.5	26.0	62.2	83.1	71.4	46.5	11.1	48.0	72.8	59.3	34.8	3.4	42.4	72.0	47.6
Tenure Bin	2-5	26.6	42.7	26.0	11.0	20.9	20.1	21.5	20.5	11.4	17.4	17.1	11.7	16.8	10.1	13.2
Tenure Bin	6+	12.9	31.2	11.8	5.9	7.7	33.5	67.5	31.5	15.8	23.3	48.1	84.9	40.8	17.9	39.2
Census Division	Unknown Census division	0.5	0.6	0.5	0.3	0.7	0.4	0.3	0.5	0.2	0.7	0.3	0.5	0.9	0.8	0.0
Census Division	New England	3.2	6.8	3.1	0.8	1.7	3.8	5.0	3.9	1.4	2.8	5.0	8.2	5.0	2.6	5.7
Census Division	Middle Atlantic	11.0	17.9	10.5	4.6	9.2	11.7	19.5	13.8	4.2	8.1	12.7	24.8	17.3	5.5	9.5
Census Division	East North Central	15.4	21.3	14.7	12.4	16.0	15.5	23.5	15.0	9.5	15.2	15.5	23.2	16.8	11.9	14.1
Census Division	West North Central	9.1	7.6	11.4	10.9	7.4	8.5	8.6	8.3	13.2	10.0	9.5	3.6	12.0	18.3	11.9
Census Division	South Atlantic	23.9	22.2	22.6	23.0	23.4	23.5	17.0	24.4	22.5	22.1	21.5	16.6	19.0	23.5	19.7
Census Division	East South Central	8.3	9.5	7.9	6.8	10.4	8.0	9.7	6.7	6.8	7.4	7.1	10.0	6.4	3.8	3.2
Census Division	West South Central	11.2	3.6	11.0	17.6	11.9	10.3	2.8	10.3	17.2	11.7	10.3	0.9	6.4	12.7	13.6
Census Division	Mountain	4.8	4.2	3.8	4.4	3.6	5.0	4.8	5.1	4.8	5.5	5.1	5.5	5.3	3.2	7.6
Census Division	Pacific	12.5	6.4	14.6	19.3	15.7	13.3	8.8	12.2	20.0	16.4	13.0	6.8	10.9	17.8	14.6

Panel B: Annual Earnings

		Ages 22-33					Ages 34-57					Ages 58-69				
		All	Bottom 10%	Top 10%			All	Bottom 10%	Top 10%			All	Bottom 10%	Top 10%		
		Pred	Act	Pred	Act		Pred	Act	Pred	Act		Pred	Act	Pred	Act	
Education	Less than HS	14.7	5.4	14.7	26.9	25.3	18.6	7.6	17.1	34.5	24.0	22.5	6.9	18.0	40.4	24.9
Education	High School Graduate	48.2	29.3	44.0	57.0	51.7	40.6	22.0	40.2	49.4	44.3	32.8	19.5	30.5	36.6	25.6
Education	Some College	15.6	18.8	15.7	9.7	12.0	14.9	17.4	15.6	9.5	13.9	13.8	19.5	15.9	10.5	13.7
Education	Bachelor's Degree	16.4	37.3	17.6	3.8	7.5	16.5	34.1	18.0	4.2	11.4	17.5	31.1	15.3	5.9	21.1
Education	Bachelors+	5.0	9.2	8.1	2.6	3.6	9.4	18.9	9.1	2.3	6.5	13.3	22.9	20.2	6.6	14.7
Race	White	58.7	85.5	57.5	25.3	42.5	60.5	86.5	60.0	31.2	47.5	69.2	86.3	77.9	54.7	67.9
Race	Black	31.5	10.2	30.3	67.5	48.8	28.1	7.5	29.7	59.1	40.0	20.6	10.5	15.8	35.4	20.5
Race	Native American	1.9	0.7	2.0	1.6	1.9	2.1	0.4	1.6	2.1	2.4	2.3	0.0	3.2	3.1	3.2
Race	Asian or Pacific Islander	0.7	1.0	0.8	1.1	1.0	1.3	1.6	0.8	0.8	1.2	1.8	1.1	0.5	1.0	1.1
Race	Latino	6.7	2.7	8.8	4.2	5.7	7.4	4.1	7.6	6.4	8.5	5.0	2.1	2.6	5.7	5.8
Race	Other or Unknown	0.5	0.0	0.5	0.3	0.1	0.7	0.0	0.4	0.4	0.5	1.0	0.0	0.0	0.0	1.6
Cohort	born pre-1944	4.6	13.6	5.6	0.8	2.2	24.0	34.9	19.5	22.4	17.5	50.8	25.3	46.3	67.7	55.3
Cohort	born 1944-1952	19.0	35.1	14.5	13.0	14.5	18.9	6.3	17.6	19.2	18.6	26.2	20.0	29.5	21.9	28.9
Cohort	born 1953-1960	22.3	14.9	20.2	31.2	25.0	20.8	3.0	21.6	25.9	26.8	21.2	48.9	22.1	9.9	15.3
Cohort	born post-1960	54.1	36.4	59.6	54.9	58.3	36.4	55.8	41.3	32.5	37.1	1.9	5.8	2.1	0.5	0.5
Tenure Bin	0-1	60.8	21.6	64.8	93.1	79.6	46.5	8.4	50.1	80.3	64.9	33.1	0.5	39.8	72.6	48.3
Tenure Bin	2-5	26.4	44.0	24.3	5.3	15.2	20.1	18.1	19.7	10.2	16.2	18.1	11.1	19.2	12.9	19.4
Tenure Bin	6+	12.8	34.4	10.9	1.7	5.2	33.4	73.5	30.2	9.5	18.8	48.7	88.4	41.1	14.5	32.2
Census Division	Unknown Census division	0.5	0.7	0.5	0.0	0.4	0.5	0.5	0.3	0.3	0.7	0.3	0.0	1.2	0.5	0.0
Census Division	New England	3.2	6.7	2.9	1.5	1.5	3.8	5.8	3.6	1.7	2.5	5.1	5.8	5.0	1.6	4.7
Census Division	Middle Atlantic	11.0	16.1	11.1	7.4	9.1	11.7	17.3	12.2	7.6	8.9	12.6	23.3	12.1	5.7	8.4
Census Division	East North Central	15.5	16.7	15.2	19.0	18.9	15.5	19.7	15.9	15.9	14.9	15.2	23.1	14.9	16.5	18.2
Census Division	West North Central	9.1	8.7	7.1	6.9	7.7	8.6	8.2	7.3	10.4	8.8	9.2	3.2	10.9	13.8	12.1
Census Division	South Atlantic	24.0	23.3	24.9	29.5	24.8	23.4	19.4	22.6	26.2	24.7	21.5	20.0	23.6	25.5	21.7
Census Division	East South Central	8.4	8.5	8.6	8.7	8.3	8.0	8.6	8.0	9.5	8.2	6.9	8.9	4.5	7.1	5.3
Census Division	West South Central	11.1	5.6	11.1	10.7	11.2	10.3	6.0	12.7	11.6	12.1	10.2	5.8	7.2	10.4	13.2
Census Division	Mountain	4.8	6.6	5.4	2.7	3.4	5.0	7.0	4.6	3.1	4.5	5.6	4.7	5.4	4.2	4.1
Census Division	Pacific	12.5	7.0	13.4	13.7	14.7	13.3	7.6	12.7	13.8	14.8	13.2	5.3	15.1	14.6	12.3

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A Appendix

Table 17: Gamma Regressions: Industry Selection Results

Hourly Earnings				Annual Earnings			
Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Public Administration	3	1	7	Not Applicable	1	1	1
Educ./Sport	11	2	18	Other Services	2	2	2
Mechanical Eng	10	3	9	Agric./Forestry	4	3	3
Not Applicable	1	4	1	Wood/Paper/Print	6	4	9
Constr. Relate	6	5	10	Public Administration	3	5	24
Other Trans.	28	6	13	Retail	20	6	8
Agric./Forestry	7	7	3	Constr. Relate	7	7	6
Retail	4	8	4	Legal Services	9	8	4
Energy/Water	15	9	28	Wholesale	30	9	12
Iron/Steel	24	10	23	Construction	12	10	5
Clothing/Text	12	11	30	Clothing/Text	14	11	31
Wholesale	13	12	12	Energy/Water	8	12	22
Postal System	22	13	19	Mechanical Eng	23	13	13
Health Service	30	14	15	Service Indust	17	14	10
Construction	5	15	6	Food Industry	19	15	18
Food Industry	26	16	27	Financial Inst	13	16	14
Electrical Eng	17	17	21	Electrical Eng	21	17	11
Other Services	2	18	2	Educ./Sport	11	18	16
Volunt./Church	16	19	20	Mining	26	19	28
Insurance	8	20	22	Chemicals	27	20	21
Chemicals	20	21	24	Health Service	5	21	25
Legal Services	31	22	8	Volunt./Church	18	22	30
Restaurants	29	23	25	Postal System	22	23	20
Earth/Clay/Stone	19	24	31	Other Trans.	10	24	7
Wood/Paper/Print	25	25	11	Insurance	16	25	23
Mining	14	26	16	Restaurants	24	26	15
Financial Inst	9	27	5	Train System	31	27	27
Train System	27	28	32	Iron/Steel	29	28	19
Social Security	33	29	17	Synthetics	15	29	32
Service Indust	18	30	14	Earth/Clay/Stone	28	30	17
Synthetics	23	31	26	Social Security	33	31	33
Fisheries	21	32	29	Fisheries	25	32	26
Priv. Househld	32	33	33	Priv. Househld	32	33	29

Table 18: Gamma Regressions: Occupation Selection Results

Hourly Earnings

Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	1	1	9
Transport. Oper	43	2	32
Priv. Bus. Leadr	2	3	2
Convey. Oper.	12	4	11
Bricklay/ Carpt	8	5	14
Labor/ Craftsmn	24	6	18
Educator	16	7	15
Inspector	30	8	40
Machine Fitter	17	9	13
Janitor	25	10	30
Dr./Dentist/Vet	3	11	1
SecurityServic	61	12	27
Pipe Fitter	75	13	39
Electr. Enginr	31	14	16
Architect/Engineer	13	15	36
Scientist	28	16	47
Vendor	5	17	4
Tool/Die Maker	21	18	59
Farm Manager	7	19	3
Eng. Tech. Expert	23	20	46
Ofc. Worker Etc	20	21	55
Mathematician	45	22	35
Painter	11	23	5
Mailman	54	24	53
Accountant	14	25	42
Legislator	4	26	7
Tailor	70	27	31
Cleric	47	28	45
Lumbrman/ Axman	10	29	12
Chemist	9	30	48
Cook/Waiter	39	31	54
Food Producer	69	32	41
Lawyer	18	33	33
Stat. Mach. Oper	35	34	57
Soldier	22	35	38
Aero/ MarineEngr	6	36	8
BusinessManagr	38	37	19
Life/Phys. Sci.	34	38	28
ComputerOperat	52	39	72
Music/Perform	50	40	51
Hair Stylist	33	41	10
Agriculturladm	26	42	37
RelatMedicalJob	19	43	17
Insurance Rep.	15	44	6
Buyer	40	45	50
ChemicalWorker	67	46	67
Service Worker	29	47	23
Bookkeepr/Cash	36	48	29
Office Manager	32	49	34
Miner	65	50	56
Author	48	51	49
Cabinet Maker	72	52	70
Prof. Athlete	37	53	20
Domestic Help	59	54	64
Sculptr/ Paintnr	49	55	22
Dry-Cleaner	60	56	65
Administrator	42	57	52
Rest./StoreMgr	41	58	25
Foundry Worker	66	59	71
Printer Etc.	27	60	24
Forestry Work	63	61	73
Tech. Salespers	56	62	58
Glazier	77	63	77
Stenographer	51	64	62
Economist	46	65	61
Transp. Attend	44	66	44
Salesperson	57	67	66
Tel. Operator	55	68	43
Shoemaker	71	69	63
Broadcaster	74	70	68
Spinner/Weaver	68	71	75
Farm Hand	62	72	21
Fisher/Hunter	64	73	26
Conductor	53	74	60
HH Supervisor	58	75	74
Stone Cutter	73	76	76
Jewelry Maker	76	77	69
Manufacturer	78	78	78

Annual Earnings

Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	6	1	1
Priv. Bus. Leadr	1	2	3
Farm Manager	5	3	2
Machine Fitter	17	4	9
Transport. Oper	3	5	7
Labor/ Craftsmn	2	6	4
Painter	14	7	8
Electr. Enginr	9	8	12
Insurance Rep.	10	9	6
Convey. Oper.	77	10	24
Bricklay/ Carpt	36	11	10
Janitor	24	12	16
Vendor	55	13	19
Architect/Engineer	8	14	40
Pipe Fitter	73	15	26
Dr./Dentist/Vet	12	16	5
Ofc. Worker Etc	54	17	32
RelatMedicalJob	30	18	44
Cook/Waiter	18	19	21
Educator	4	20	14
Printer Etc.	33	21	17
Rest./StoreMgr	22	22	42
Bookkeepr/Cash	15	23	11
Inspector	21	24	37
BusinessManagr	20	25	15
Eng. Tech. Expert	39	26	39
SecurityServic	19	27	18
Tool/Die Maker	7	28	13
Chemist	25	29	62
Sculptr/ Paintnr	32	30	36
Mathematician	42	31	22
Legislator	48	32	25
Economist	26	33	38
Author	45	34	47
Service Worker	59	35	49
Life/Phys. Sci.	41	36	53
Domestic Help	28	37	30
Administrator	49	38	43
Accountant	23	39	52
Tailor	31	40	46
Soldier	78	41	45
Food Producer	68	42	50
Music/Perform	35	43	29
Spinner/Weaver	67	44	69
Scientist	47	45	51
Lawyer	43	46	35
ChemicalWorker	66	47	23
Office Manager	27	48	27
Lumbrman/ Axman	65	49	20
Mailman	13	50	54
Buyer	29	51	41
Aero/ MarineEngr	40	52	57
Cleric	44	53	48
Stenographer	50	54	70
ComputerOperat	51	55	61
Stat. Mach. Oper	37	56	60
Foundry Worker	64	57	63
Forestry Work	61	58	71
Dry-Cleaner	11	59	33
Cabinet Maker	70	60	58
Agriculturladm	60	61	56
Tel. Operator	16	62	31
Broadcaster	72	63	66
Hair Stylist	58	64	59
Miner	63	65	67
Farm Hand	34	66	28
Salesperson	56	67	64
Jewelry Maker	75	68	76
Glazier	74	69	77
Prof. Athlete	46	70	65
Conductor	53	71	72
Transp. Attend	52	72	55
Shoemaker	69	73	74
Manufacturer	76	74	78
Tech. Salespers	38	75	34
HH Supervisor	57	76	73
Fisher/Hunter	62	77	68
Stone Cutter	71	78	75

Table 19: Alpha Regressions: Industry Selection Results

Hourly Earnings

Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Public Administration	8	1	9
Not Applicable	1	2	1
Mechanical Eng	11	3	11
Other Services	2	4	2
Agric./Forestry	7	5	5
Chemicals	12	6	32
Postal System	26	7	24
Educ./Sport	13	8	18
Electrical Eng	22	9	20
Wood/Paper/Print	17	10	15
Energy/Water	25	11	17
Health Service	18	12	19
Constr. Relate	3	13	3
Food Industry	19	14	26
Iron/Steel	20	15	23
Legal Services	6	16	6
Synthetics	21	17	33
Clothing/Text	29	18	30
Insurance	15	19	28
Other Trans.	9	20	10
Wholesale	14	21	13
Earth/Clay/Stone	30	22	22
Volunt./Church	23	23	27
Construction	5	24	4
Train System	28	25	29
Service Indust	10	26	8
Financial Inst	31	27	16
Retail	4	28	7
Social Security	33	29	25
Fisheries	27	30	21
Mining	24	31	14
Restaurants	16	32	12
Priv. Household	32	33	31

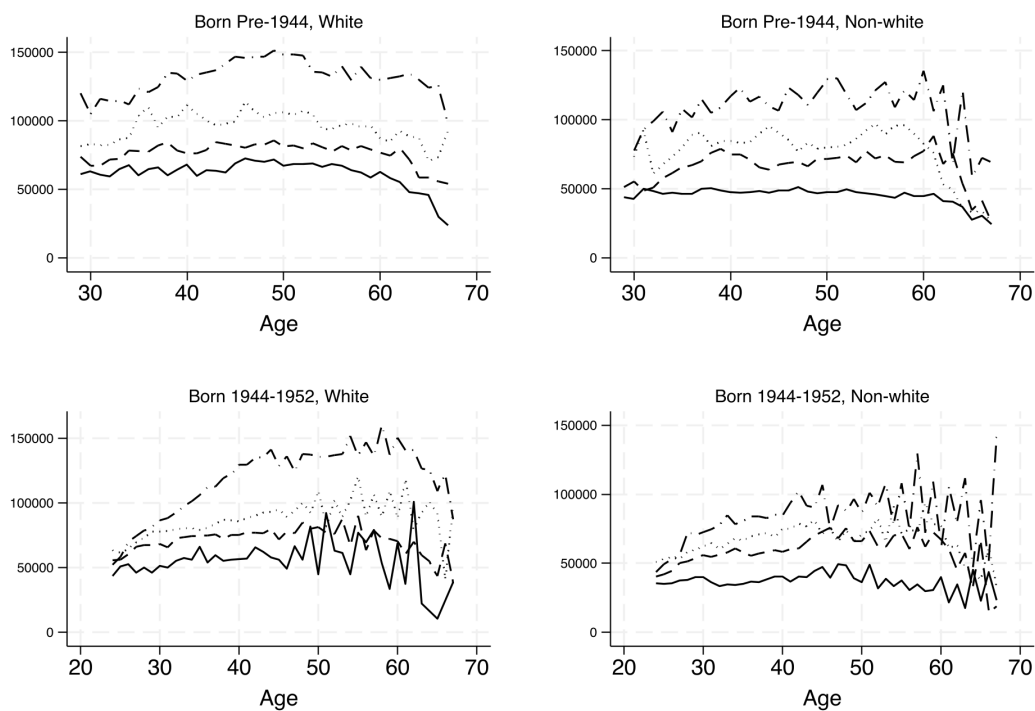
Annual Earnings

Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	1	1	1
Public Administration	5	2	8
Other Services	2	3	2
Educ./Sport	8	4	20
Wood/Paper/Print	23	5	14
Agric./Forestry	4	6	5
Mechanical Eng	26	7	13
Energy/Water	14	8	24
Health Service	25	9	17
Electrical Eng	17	10	29
Postal System	10	11	26
Retail	6	12	6
Other Trans.	16	13	10
Iron/Steel	18	14	19
Wholesale	22	15	11
Clothing/Text	11	16	31
Chemicals	13	17	28
Food Industry	19	18	23
Insurance	21	19	21
Synthetics	20	20	25
Financial Inst	28	21	15
Constr. Relate	3	22	3
Mining	24	23	22
Legal Services	7	24	4
Volunt./Church	31	25	27
Construction	9	26	7
Train System	27	27	16
Service Indust	15	28	12
Earth/Clay/Stone	30	29	18
Social Security	33	30	30
Restaurants	12	31	9
Fisheries	29	32	33
Priv. Household	32	33	32

Table 20: Alpha Regressions: Occupation Selection Results

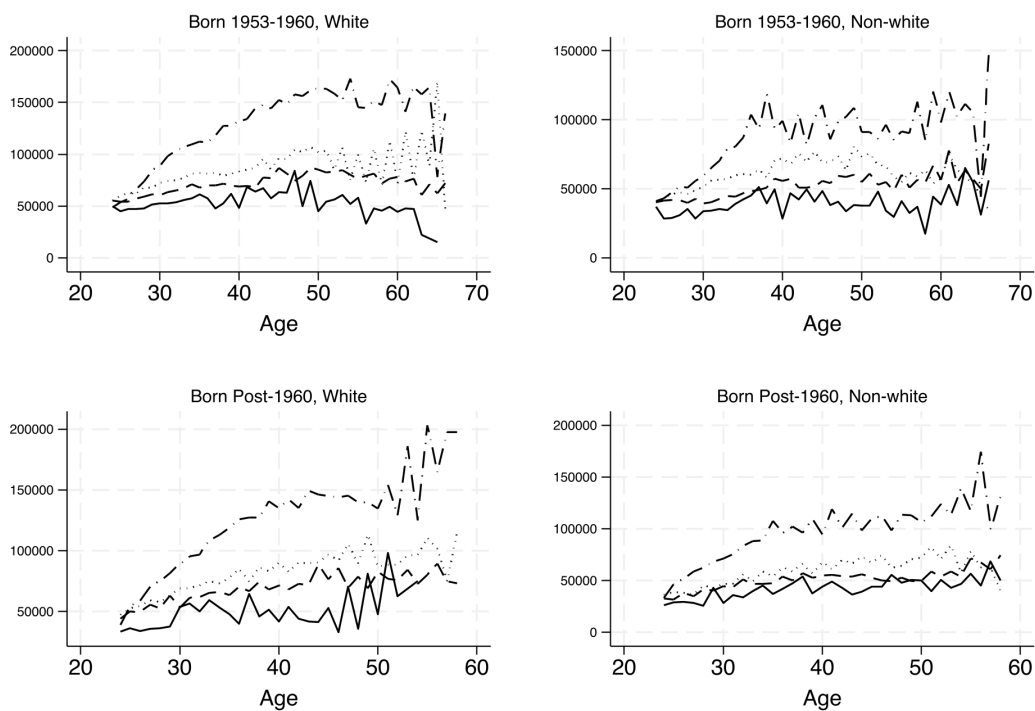
Hourly Earnings				Annual Earnings			
Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	1	1	1	Not Applicable	1	1	1
Machine Fitter	12	2	10	Priv.Bus.Leadr	3	2	5
Mathematician	4	3	34	Mathematician	4	3	40
Inspector	10	4	47	Architect/Engineer	8	4	32
Priv.Bus.Leadr	2	5	3	Machine Fitter	13	5	9
Architect/Engineer	7	6	13	Inspector	11	6	43
Ofc.Worker Etc	14	7	50	Farm Manager	2	7	2
Farm Manager	3	8	2	Ofc.Worker Etc	6	8	25
Convey. Oper.	5	9	29	Eng.Tech.Expert	16	9	62
Electr. Enginr	21	10	16	Transport.Oper	10	10	3
Mailman	9	11	55	SecurityServic	30	11	20
SecurityServic	27	12	39	Scientist	21	12	47
Labor/Craftsmn	39	13	8	BusinessManagr	17	13	31
Janitor	16	14	23	Accountant	15	14	29
Accountant	31	15	22	Educator	22	15	33
Scientist	28	16	18	Vendor	20	16	18
BusinessManagr	11	17	51	Electr. Enginr	72	17	15
Transport.Oper	18	18	5	Mailman	24	18	24
Eng.Tech.Expert	17	19	30	Lawyer	55	19	56
Insurance Rep.	6	20	4	Food Producer	18	20	51
Soldier	20	21	21	Soldier	36	21	45
Stat.Mach.Oper	35	22	70	Tool/Die Maker	40	22	44
Pipe Fitter	76	23	20	Office Manager	26	23	53
RelatMedicalJob	56	24	38	Administrator	58	24	63
Food Producer	23	25	40	Convey. Oper.	78	25	8
Bricklay/Carpt	22	26	6	Insurance Rep.	5	26	7
Printer Etc.	41	27	37	Aero/MarineEngr	39	27	65
Educator	8	28	19	Bricklay/Carpt	9	28	4
Bookkeepr/Cash	47	29	28	ComputerOperat	31	29	67
Office Manager	42	30	53	Janitor	29	30	19
Cook/Waiter	13	31	7	ChemicalWorker	43	31	64
Lawyer	37	32	35	Labor/Craftsmn	12	32	6
Tool/Die Maker	30	33	64	Printer Etc.	50	33	36
Foundry Worker	71	34	65	Lumbrman/Axman	7	34	12
Cleric	58	35	62	Dr./Dentist/Vet	28	35	46
Vendor	45	36	17	Author	57	36	42
Aero/MarineEngr	15	37	49	Bookkeepr/Cash	59	37	23
ComputerOperat	34	38	72	Pipe Fitter	14	38	11
Administrator	61	39	52	Music/Perform	35	39	21
Lumbrman/Axman	33	40	11	Stat.Mach.Oper	77	40	60
ChemicalWorker	72	41	69	Painter	23	41	10
Stenographer	51	42	63	Cook/Waiter	37	42	13
Music/Perform	29	43	9	Spinner/Weaver	45	43	68
Agriculturladm	40	44	60	Life/Phys. Sci.	53	44	58
Tailor	43	45	33	Legislator	19	45	16
Legislator	32	46	15	Tailor	69	46	39
Hair Stylist	38	47	12	Cleric	56	47	37
Rest./StoreMgr	19	48	26	Economist	54	48	61
Spinner/Weaver	48	49	59	RelatMedicalJob	25	49	22
Broadcaster	75	50	41	Fisher/Hunter	41	50	35
Forestry Work	55	51	66	Sculptr/Paintr	27	51	14
Painter	46	52	27	Stenographer	51	52	70
Sculptr/Paintr	60	53	24	Buyer	62	53	57
Author	59	54	46	Domestic Help	65	54	55
Tel. Operator	64	55	48	Hair Stylist	32	55	30
Glazier	53	56	77	Rest./StoreMgr	44	56	26
Dr./Dentist/Vet	26	57	32	Chemist	52	57	54
Economist	57	58	44	Dry-Cleaner	47	58	34
Cabinet Maker	36	59	54	Tech.Salespers	33	59	28
Fisher/Hunter	25	60	14	Glazier	75	60	77
Miner	50	61	57	Tel. Operator	48	61	48
Dry-Cleaner	68	62	45	Miner	68	62	50
Service Worker	69	63	42	Transp. Attend	60	63	71
Life/Phys. Sci.	52	64	25	Salesperson	63	64	66
Prof. Athlete	44	65	43	Foundry Worker	34	65	41
Domestic Help	49	66	31	Service Worker	42	66	17
Jewelry Maker	77	67	76	Conductor	61	67	72
Chemist	24	68	36	Agriculturladm	38	68	27
Shoemaker	73	69	71	Broadcaster	73	69	69
Buyer	65	70	58	Shoemaker	49	70	49
Salesperson	66	71	68	Forestry Work	67	71	74
Tech.Salespers	54	72	56	Cabinet Maker	70	72	52
Stone Cutter	74	73	75	Prof. Athlete	46	73	38
Farm Hand	70	74	61	Jewelry Maker	74	74	76
Transp. Attend	62	75	67	Farm Hand	66	75	59
Conductor	63	76	73	Manufacturer	76	76	78
HH Supervisor	67	77	74	HH Supervisor	64	77	73
Manufacturer	78	78	78	Stone Cutter	71	78	75

Figure 5: Average Annual Earnings by Age, Born Pre-1944 & 1944-1952



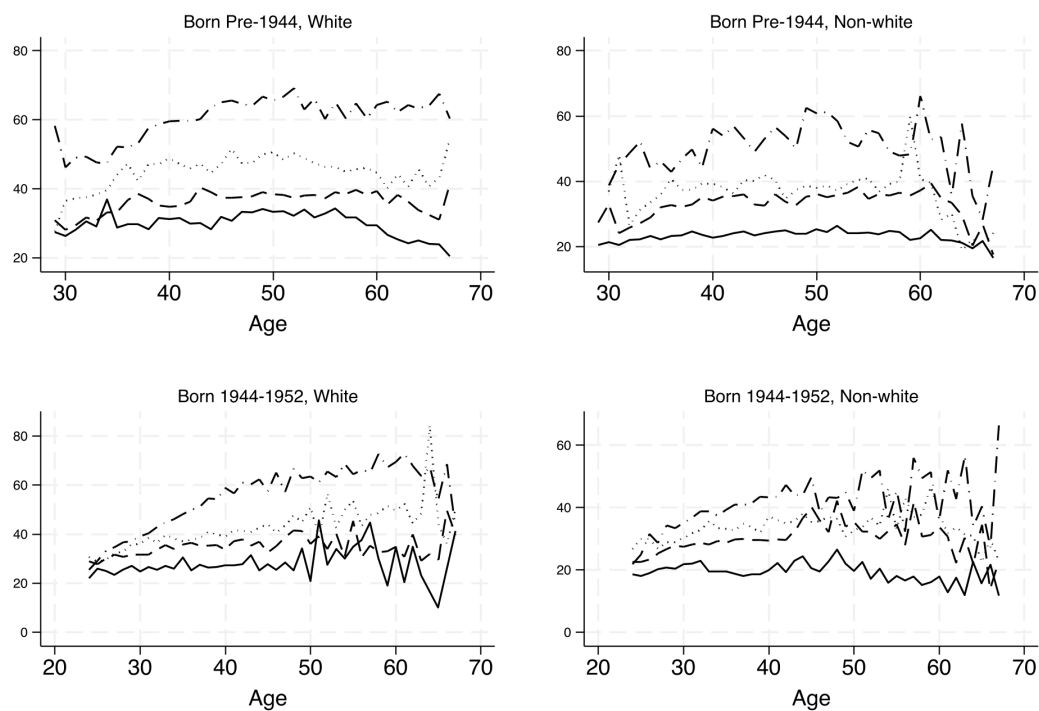
Solid = HS Dropout; Dashed = HS Graduate; Dotted = Some College; Dash-dot = College Graduate.
Annual earnings in 2024 dollars.

Figure 6: Average Annual Earnings by Age, Born 1953-1960 & Post-1960



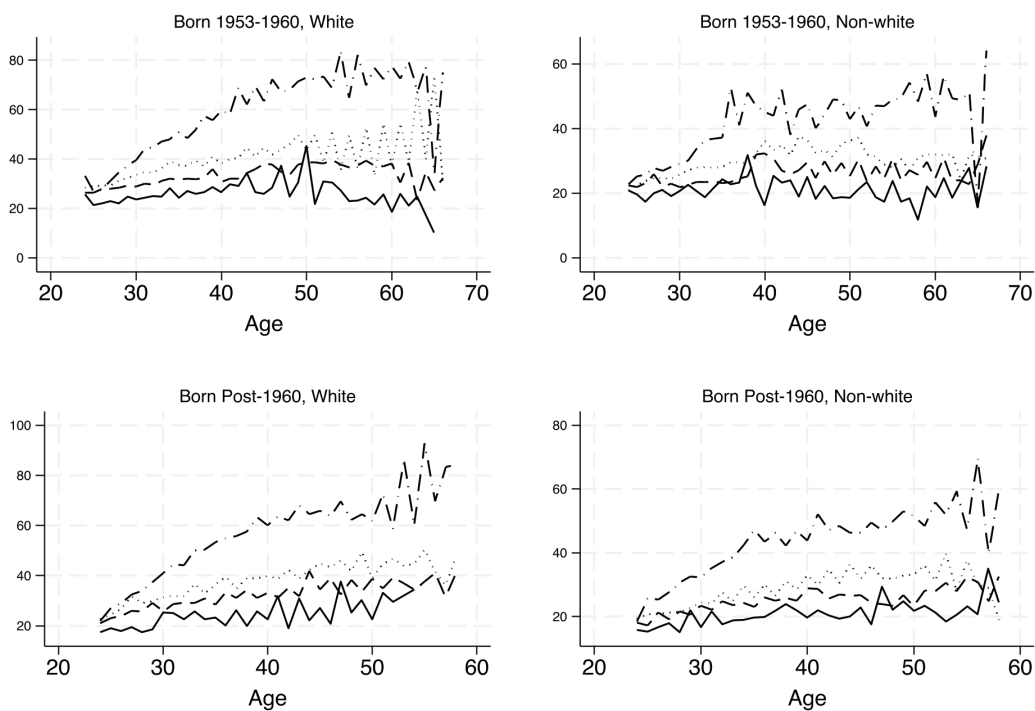
Solid = HS Dropout; Dashed = HS Graduate; Dotted = Some College; Dash-dot = College Graduate.
Annual earnings in 2024 dollars.

Figure 7: Average Hourly Wage by Age, Born Pre-1944 & 1944-1952



Solid = HS Dropout; Dashed = HS Graduate; Dotted = Some College; Dash-dot = College Graduate.
Hourly wage in 2024 dollars.

Figure 8: Average Hourly Wage by Age, Born 1953-1960 & Post-1960



Solid = HS Dropout; Dashed = HS Graduate; Dotted = Some College; Dash-dot = College Graduate.
Hourly wage in 2024 dollars.

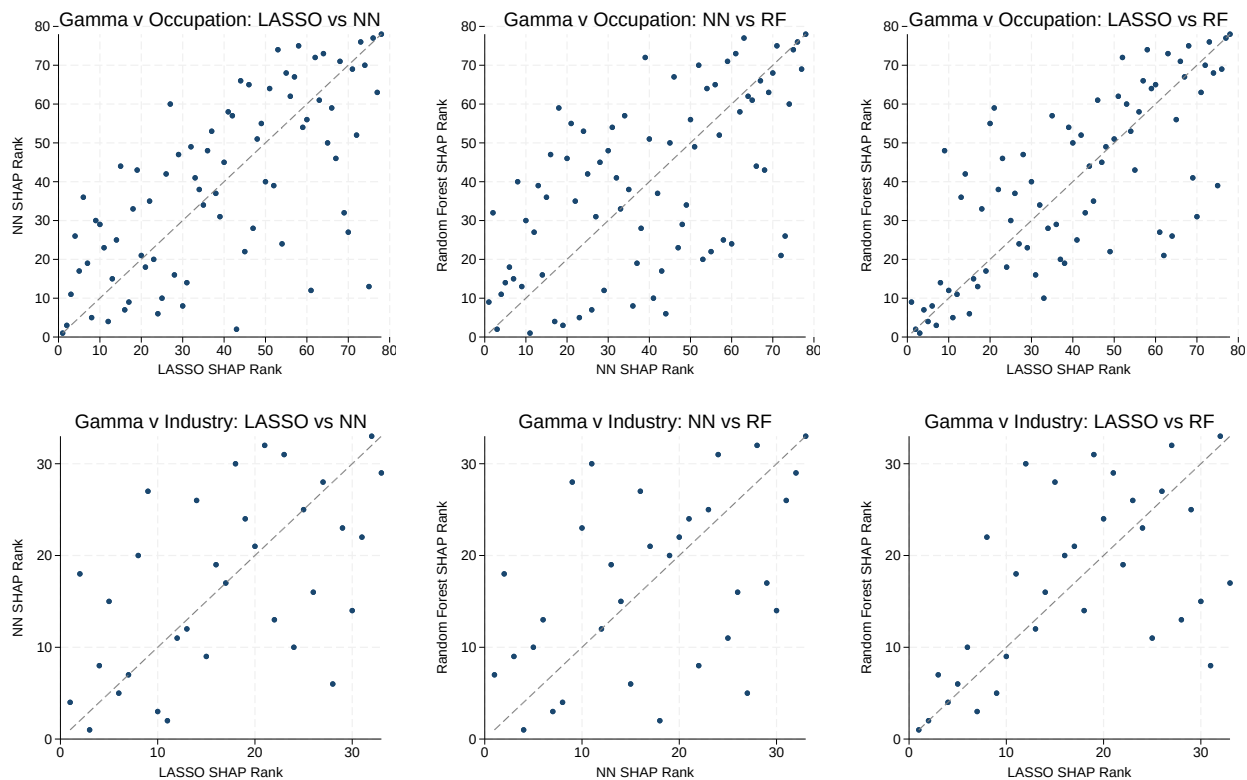


Figure 9: Gamma SHAP Rank Comparisons Across Methods, Hourly Earnings (flwage). Top row: occupation; bottom row: industry.

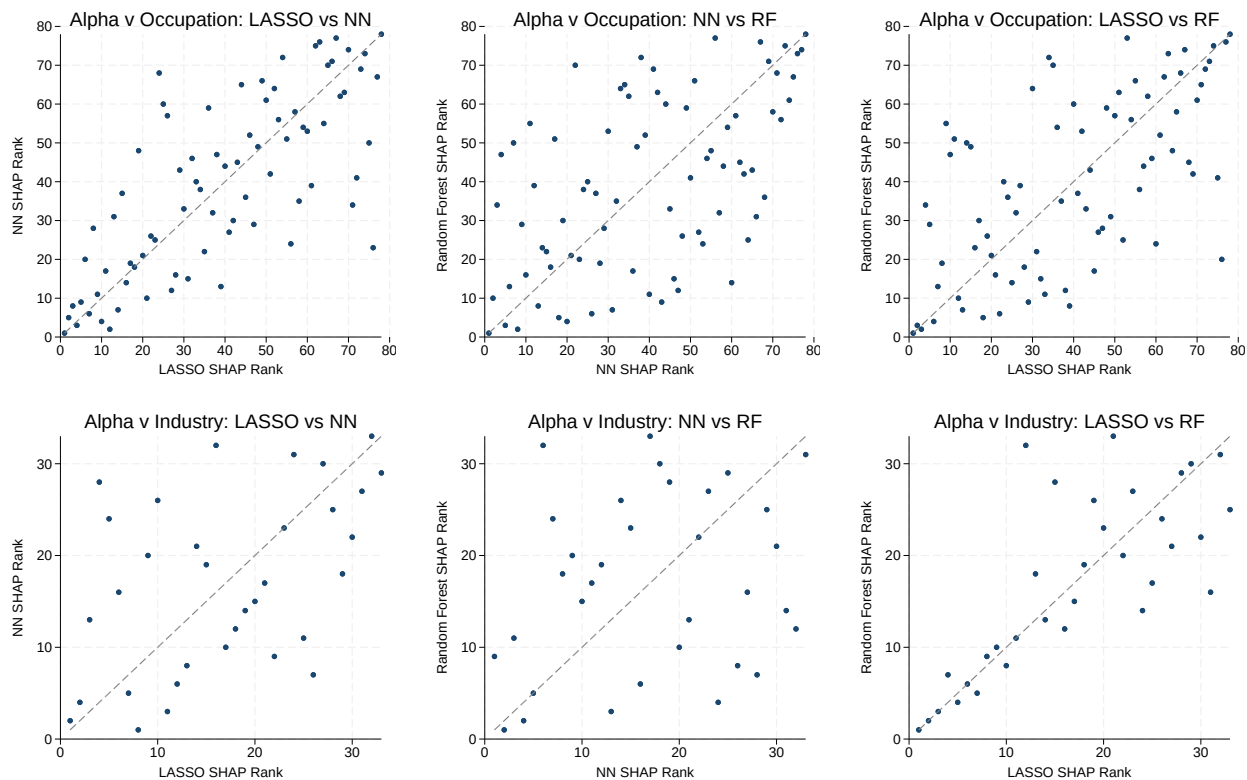


Figure 10: Alpha SHAP Rank Comparisons Across Methods, Hourly Earnings (flwage). Top row: occupation; bottom row: industry.

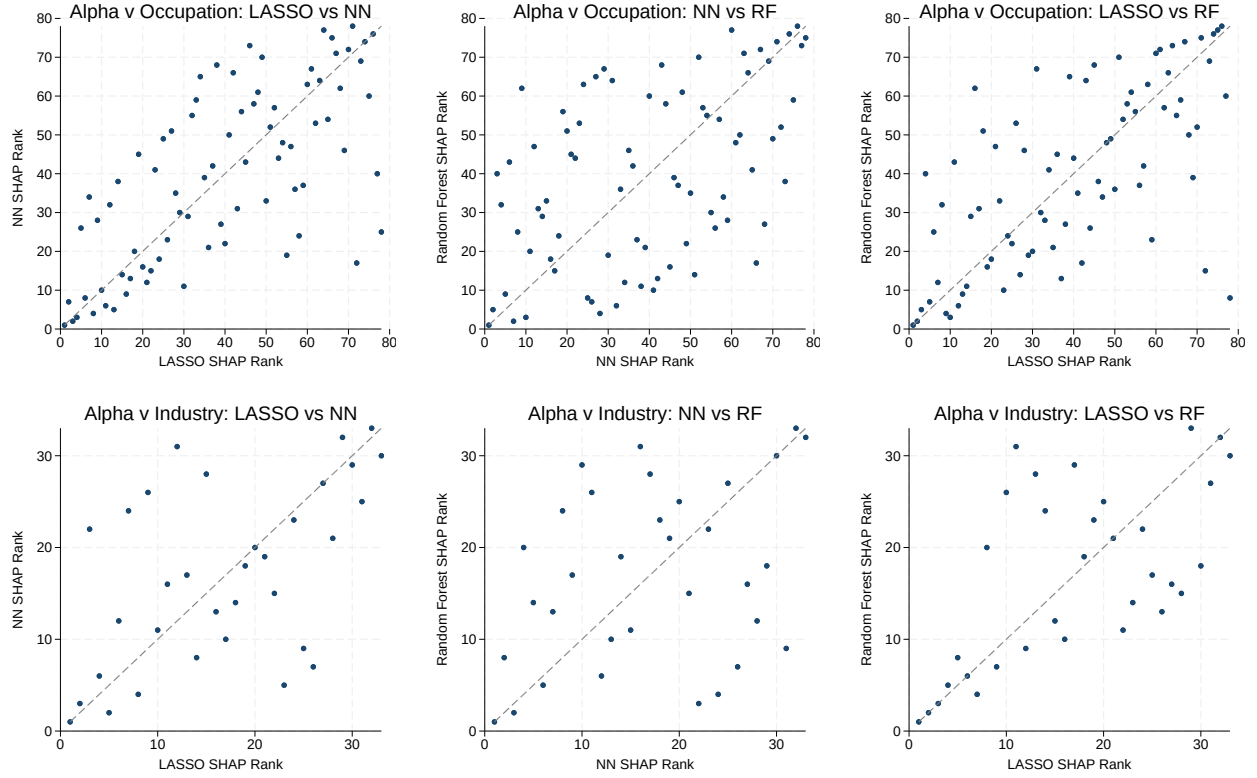


Figure 11: Alpha SHAP Rank Comparisons Across Methods, Annual Earnings (fearn). Top row: occupation; bottom row: industry.

Table 21: Group SHAP Shares of Controls, Gamma (Average —SHAP— per Variable x 100)

	Hourly Wage			Annual Earnings		
	NN	RF	LASSO	NN	RF	LASSO
Census Division	0.0001	0.0057	0.0073	0.0030	0.0087	0.0115
Year	0.0000	0.0023	0.0053	0.0006	0.0045	0.0072
Race	0.0002	0.0047	0.0229	0.0026	0.0084	0.0110
Cohort	0.0004	0.0128	0.0209	0.0048	0.0091	0.0144
Occupation	0.0000	0.0018	0.0028	0.0003	0.0021	0.0024
Industry	0.0000	0.0029	0.0066	0.0008	0.0057	0.0085
Education	0.0003	0.0105	0.0467	0.0045	0.0112	0.0110
Age	0.0003	0.0267	0.0219	0.0039	0.0330	0.0127
Tenure	0.0001	0.0311	0.0038	0.0006	0.1073	0.0257

Table 22: Mean Predicted Gamma by Education and Age Bin, Hourly Earnings

Actual							Neural Network						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0149	0.0167	0.0169	0.0118	0.0116	0.0359	Less than HS	0.0186	0.0186	0.0186	0.0186	0.0186	0.0187
High School Graduate	0.0152	0.0149	0.0139	0.0154	0.0139	0.0228	High School Graduate	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
Some College	0.0189	0.0189	0.0138	0.0201	0.0114	0.0462	Some College	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186
Bachelor's Degree	0.0222	0.0175	0.0160	0.0174	0.0236	0.0006	Bachelor's Degree	0.0186	0.0186	0.0186	0.0186	0.0186	0.0187
Bachelors+	0.0287	0.0193	0.0114	0.0135	0.0207	0.0203	Bachelors+	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186

Random Forest							LASSO						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0151	0.0153	0.0144	0.0134	0.0140	0.0271	Less than HS	0.0149	0.0150	0.0130	0.0141	0.0133	0.0231
High School Graduate	0.0155	0.0152	0.0138	0.0139	0.0147	0.0237	High School Graduate	0.0156	0.0149	0.0126	0.0144	0.0138	0.0225
Some College	0.0166	0.0164	0.0147	0.0156	0.0161	0.0280	Some College	0.0189	0.0178	0.0150	0.0169	0.0165	0.0252
Bachelor's Degree	0.0189	0.0172	0.0156	0.0154	0.0191	0.0160	Bachelor's Degree	0.0187	0.0183	0.0160	0.0175	0.0176	0.0273
Bachelors+	0.0265	0.0182	0.0134	0.0134	0.0188	0.0270	Bachelors+	0.0193	0.0177	0.0138	0.0161	0.0163	0.0248

Table 23: Mean Predicted Gamma by Education and Age Bin, Annual Earnings

Actual							Neural Network						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0294	0.0296	0.0265	0.0236	0.0198	0.0308	Less than HS	0.0280	0.0281	0.0275	0.0269	0.0274	0.0283
High School Graduate	0.0267	0.0256	0.0204	0.0200	0.0237	0.0513	High School Graduate	0.0281	0.0282	0.0276	0.0272	0.0275	0.0282
Some College	0.0272	0.0261	0.0254	0.0284	0.0300	0.0695	Some College	0.0284	0.0285	0.0280	0.0275	0.0279	0.0286
Bachelor's Degree	0.0266	0.0205	0.0208	0.0231	0.0415	0.0286	Bachelor's Degree	0.0280	0.0281	0.0276	0.0272	0.0276	0.0282
Bachelors+	0.0296	0.0241	0.0213	0.0158	0.0289	0.0688	Bachelors+	0.0280	0.0281	0.0276	0.0272	0.0275	0.0282

Random Forest							LASSO						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0311	0.0278	0.0243	0.0226	0.0236	0.0341	Less than HS	0.0297	0.0283	0.0249	0.0235	0.0285	0.0495
High School Graduate	0.0276	0.0248	0.0220	0.0211	0.0252	0.0395	High School Graduate	0.0268	0.0249	0.0216	0.0208	0.0251	0.0420
Some College	0.0290	0.0266	0.0247	0.0239	0.0290	0.0516	Some College	0.0312	0.0290	0.0258	0.0254	0.0305	0.0489
Bachelor's Degree	0.0279	0.0241	0.0214	0.0222	0.0327	0.0529	Bachelor's Degree	0.0260	0.0250	0.0221	0.0214	0.0271	0.0429
Bachelors+	0.0315	0.0255	0.0207	0.0195	0.0302	0.0705	Bachelors+	0.0265	0.0250	0.0212	0.0213	0.0265	0.0434